

LANKA MOKSHAGNA SRI SAI

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PROFILE SUMMARY

Passionate about technology and intelligent system design, with hands-on experience in Embedded Systems, Neuromorphic AI, and Full-Stack Development. Enjoys bridging the gap between hardware and software by building smart, cyber-physical solutions for real-world challenges. Interested in opportunities that foster innovation and enable the application of technical and analytical skills[click here](#).

EDUCATION

2023 – Present	B.Tech in Electronics and Communication Engineering Amrita Vishwa Vidyapeetham, Bengaluru	(CGPA: 6.32)
2021 – 2023	Class 12th – Ascent Junior College, Visakhapatnam, Andhra Pradesh	(74.2%)
2019 – 2021	Class 10th – Tirumala High School, Visakhapatnam, Andhra Pradesh	(100%)

PROJECTS

Multimodal Osteoporosis Screening System Sep 2025 – Present

- Engineered a cyber-physical system integrating Ultrasound, RF, and Mechanical Impulse sensing using a Spiking Neural Network for intelligent diagnostic analysis.
- Developed a high-speed DAQ bridge (Arduino Nano, 115200 baud) streaming real-time data to a Python GUI with Explainable AI and QR-based medical reporting.

Morse Code Generator using LPC2148 May 2025

- Implemented a real-time Morse code generator using LPC2148 (ARM7) with UART-based alphanumeric input, lookup table translation, and 16x2 LCD visual feedback.
- Configured Timer0 for precise 1kHz tone generation, producing standardized Morse timings (100ms dot, 300ms dash).

Advanced Underwater Image Enhancement & Object Detection Framework Oct 2025 – Present

- Designed a multi-stage physics-based enhancement pipeline (Fourier filtering + DCP dehazing + Laplace decomposition) to mitigate underwater light attenuation and scattering.
- Developed an object detection and classification pipeline using YOLOv2 and a custom 10-layer CNN that achieved 96.46

SKILLS

VLSI Design	Basics of Verilog
FPGA & Tools	Xilinx Vivado, MATLAB
Microcontrollers	Keil µVision, LPC2148, Proteus
Programming Languages	C (Basics), Embedded C, Python, JavaScript, SQL, HTML/CSS
Foundational Knowledge	Digital Electronics, CMOS Logic, Analog Electronics, Communication Systems